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Nitrogen Injection Application for Oil Recovery in Trinidad

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Abstract

Improved oil recovery (IOR) has been attracting renewed interest amongst many operators in Trinidad in recent times. High oil prices and government incentives have increased number of studies and pilot projects in secondary and tertiary recovery processes. Whilst IOR selection is driven by rock and fluid conditions, historically most operators still gave preference to infill and outstep drilling campaigns for that last hope of primary oil. Reviews conducted on Trinidad's past projects indicated mixed economic success of IOR projects with thermal flood projects being most successful.

Nitrogen injection has never been extensively analyzed or implemented in Trinidad and now more than before is the most opportune period to establish its place amongst the other injection mediums. This paper reviews the history and use of nitrogen injection around the globe, focuses on application to Trinidad's fields and analyses the reservoir/geologic parameter impact on recovery through reservoir simulation. Numerical simulation was performed on blocks within an onshore mature field and recovery results showed promising project potential. Simulation was also carried out on synthetic models which mimicked the various structural dips and porosity-permeability relationships that Trinidad's southern onshore acreage possesses. The findings herein, based on immiscible injection dynamics, explains the mechanism of nitrogen injection for Trinidad's shallow onshore reservoirs.

The overall results suggest that nitrogen injection is a real alternative and much more practical and economic than previously envisioned by investors. The technology of modern day reservoir simulation has made analysis fast and efficient using structural, stratigraphic and reservoir properties that reflects the subsurface in Trinidad. The encouraging results of this study means nitrogen injection for enhancing oil recovery can now be seriously considered by mature acreage operators who wish to lengthen the economic life span of their fields.

Introduction

Trinidad & Tobago's onshore acreage has seen many stages of development since this country's oil industry began, over a century ago. As a result, many of Trinidad's onshore fields are in a mature stage of development, with many of them operating under various forms of artificial lift. Several of these mature fields have been leased out to private investors/operators by the state since continued production from these fields has been deemed uneconomical. Many of these fields have the potential to become once again economically viable, once suitable production optimization strategies and techniques are implemented.

Optimization strategies, whilst necessary have become ineffective in adding significant reserves and in increasing production rates; a more robust approach, such as enhanced oil recovery (EOR) needs to be considered to fully exploit the vast quantities of hydrocarbons still trapped in these mature fields.

EOR is not a new concept or development to Trinidad & Tobago. In fact, Trinidad and Tobago has implemented many EOR projects in the past. These include immiscible carbon dioxide floods, steamfloods, cyclic steam stimulation and microbial enhanced oil recovery (MEOR) methods. The results of the various attempted EOR projects have been promising and indicate that EOR has an important role in the future development and production of Trinidad's hydrocarbon resources.

Trinidad and Tobago first introduced EOR in 1966. The first EOR project implemented was a thermal oil recovery project consisting of a small cyclic steam stimulation pilot in the Palo Seco Field. Since then several steamfloods have been

implemented in other onshore fields such as Central Los Bajos, Guapo, North Fyzabad, Apex-Quany/Coora/Quarry and most recently in UMLE formation in Forest Reserve field.

Despite its successes, these thermal recovery projects faced several challenges. Some of these included:

- Well Damage
- Injection Supply Problems
- Gravity Override of Steam
- Surveillance Issues

Between 1973 and 2003, four (4) immiscible carbon dioxide (CO₂) pilot floods were implemented by the Petroleum Company of Trinidad and Tobago (Petrotrin) in the Forest Reserve and Oropouche fields. Three (3) floods were conducted at Forest Reserve targeting the Upper Cruse Sands, Lower Forest Sands and the Upper Forest Sands. One (1) flood was conducted in Oropouche targeting the AO 08 Sand. Despite increases in oil recovery, the projects were again faced with several challenges. Some of these included:

- Unreliable CO₂ supply. (Most significant)
- Unreliable equipment
- CO₂ breakthrough
- High GOR (pattern issues)
- Poor Sweep Efficiency (pattern issues)
- Limited Pressure and Gas Measurements
- Corrosion.

The problems highlighted in past EOR projects have prompted many local operators to explore new economic methods and processes to attempt to recover the large quantities of hydrocarbons still trapped in these mature fields. One method which has gained considerable attention from some local operators is the use of nitrogen injection.

The use of nitrogen injection to enhance oil recovery has been used successfully worldwide since the mid 1960's and its use is becoming increasingly popular due to its lower production costs and availability when compared to conventional hydrocarbon gases, making it a popular choice among small local field operators (Clancy et al, 1985).

This report seeks to examine the use of immiscible nitrogen injection to enhance oil recovery. It will attempt to examine, through the use of reservoir simulation, the processes occurring as a result of nitrogen injection and also to determine what are some of the important factors required for nitrogen injection a success and finally to assess the potential for nitrogen injection in other similar mature onshore fields in Trinidad and Tobago.

Interestingly, the very first reported use of nitrogen injection in reservoirs also coincided with the advent of EOR in the U.S. The origins of nitrogen injection and thus EOR in the U.S. can be traced back to the prolific Permian Basin in West Texas. In 1945 Atlantic Richfield discovered an unusual field, which they called Block 31. Block 31 contained an estimated 300 million barrels of light, gassy OOIP and had an initial reservoir pressure of approximately 4,000 psi. The development of Block 31 proved to be a challenge because of its low permeability (1 millidarcy). This resulted in very poor well production rates. In an effort to overcome this problem, studies of fluid movements within the reservoir were conducted. These results of these studies showed that injecting natural gas at pressures sufficiently high that it became miscible with oil created an oil/gas phase that was much more mobile than the oil phase alone. Thus, the oil/gas phase could permeate through the reservoir toward wellbores more easily.

In 1949, Atlantic-Richfield began compressing and injecting natural gas from a nearby source into Block 31. The field began to produce, slowly at first, but production grew steadily. By 1965 cumulative production from Block 31 approached 90 million barrels. The industry hailed the success at Block 31, recognizing that miscible gas injection could extract oil that would remain in the ground under conventional methods. With such poor permeability, no oil was extracted under primary or secondary methods; all production was tertiary or enhanced oil recovery. In 1966, to avoid using marketable natural gas, Atlantic Richfield developed a system to inject flue gas (deficient in oxygen but rich in nitrogen and carbon dioxide) from a nearby processing plant.

Production gradually increased to a peak of 20,000 BOPD in 1978. However, production slid steadily thereafter, to 2,500 BOPD in 1998, with an average production of 15 BOPD per well (Standing, 2007). The success of Block 31 led to the emergence of a number of other enhanced oil recovery projects utilizing nitrogen in Texas, Louisiana, Wyoming, Utah, Oklahoma and California. Table 1 shows the oil recovery from some of the nitrogen projects listed above in the above

locations. In more recent times newer nitrogen projects have been reported as seen in Table 2 and further to this there also exist recent developments in Canada's Turner Valley which is operated by Talisman and began in 2001 as well as in Mexico's Bay of Campeche, which contains the Cantarell-Akal field. This field is one of the largest oilfields in the world and currently has the world's largest nitrogen injection project.

Theory

The immiscible displacement of oil by gas involves injecting gas, at pressures below which miscibility occurs, into an oil reservoir into a gas cap if one exists or directly into the oil zone (Latil, 1980). It is a process in which there is no mixing of the injected gas and displaced oil phase at the pore level through mass transfer of components.

In immiscible gas injection processes, the injected gas helps improve the mobility and also help to lower the interfacial tension. However, the degree to which these occurrences take place depends on several factors including pressure, temperature and compositions of both the reservoir fluids and injected gas. Several techniques including: analytical and empirical methods as well as ternary plots have been used to describe and predict the behavior of immiscible gas injection processes.

The resulting displacement may be either miscible or immiscible, depending on the pressure, temperature and oil composition. Because of its low cost, large volumes of nitrogen maybe injected. Nitrogen can also be considered for use as a chase gas in hydrocarbon miscible or carbon dioxide floods. Nitrogen is inferior to hydrocarbon gases (and much inferior to carbon dioxide) from an oil recovery point of view for the miscible process. Nitrogen has a lower viscosity, poor solubility in oil and requires much higher pressure to generate or develop miscibility.

Nitrogen recovers oil by accomplishing the following:

- Vaporizing the lighter components of the crude oil and generating miscibility, given sufficient pressure.
- Providing a gas drive whereby a significant portion of the reservoir volume is filled with low cost gases.

The process limitations include the following:

- Developed miscibility can only be achieved with light oils and at high pressure, therefore seep reservoirs are needed. This is not the case with respect to existing well and reservoir availability onshore Trinidad.
- A steeply dipping reservoir is desired to permit gravity stabilization of the displacement, which has an unfavorable mobility ratio

Problems associated with this process include the following:

- Viscous fingering results in poor vertical and horizontal sweep efficiency.
- Nitrogen must be separated from the saleable produced gas.

Methodology

A reservoir characterization approach was adapted in studying a section of a Trinidad onshore mature field whose properties can be seen in Table 3. Detailed flow unit identification through petrophysical evaluation allowed the building of a numerical 3d model populated with well trajectories, completion and production data. The 25deg -30 deg dipping reservoir was then history matched to an acceptable stage. The study moved forward by exploring four (4) scenarios: (1) Crestal or gas cap injection, (2) Oil zone injection, (3) Water zone injection and (4) Simultaneous crestal or gas cap and water zone injection.

The real case results were analysed and based on oil recovery comparison for the above scenarios. Further to the real case study a standard 3d model was built to investigate application to other fields in Trinidad. Three parameters, which were determined to be the most important for gas injection EOR projects, were investigated to analyse their effect on recovery.

These parameters were: Dip, Crude API and Porosity/Permeability relationship. After simulation result graphs of recovery versus porosity-permeability relationship (Appendix 2) were built for the following scenarios:

1. Scenario 1: Dip of 10 degrees, 25 API crude
2. Scenario 2: Dip of 20 degrees, 25 API crude
3. Scenario 3: Dip of 30 degrees, 25 API crude
4. Scenario 4: Dip of 10 degrees, 30 API crude
5. Scenario 5: Dip of 20 degrees, 30 API crude
6. Scenario 6: Dip of 30 degrees, 30 API crude

The porosity-permeability relations used were as follows (% - md): 15% –20 md, 20% - 40 md, 25% - 100 md, 25% - 800 md, 17% - 80 md, 22% - 150 md, and 30% - 1000 md. A 25 acre rectangular model was used. The sand was 50' thick with 3 wells penetrating reservoir, two being down dip and one being up dip in gas cap. The residual oil saturation was used as 15 % and the reservoir pressure after primary production was 400 psi. No OWC was introduced in the model. An injection rate of 500 Mscf/d was used in the one updip well.

Results and Discussion

Real Case Study

Four (4) scenarios: crestal or gas cap injection, oil zone injection, water zone injection and simultaneous crestal or gas cap and water zone injection, were investigated along with the effect of several parameters including number of injection wells, injection rates, GOR and produced nitrogen. The results are summarized below in Appendix 1, Figure 1. From this it is evident, as expected, that injecting nitrogen updip into the secondary gas cap would result in maximum oil recovery.

Noteworthy comments include:

- Increasing the number of injection wells did not result in an increase in oil recovery and hence cumulative oil production because of a faster nitrogen breakthrough time
- Higher nitrogen injection rates resulted in higher initial oil rates and cumulative oil production, but lower recoveries, also due to faster nitrogen breakthrough time
- Lower injection rates resulted in higher recoveries due to a longer breakthrough time
- Higher injection rates resulted in a shorter project life, compared to lower injection rates, because of a faster nitrogen breakthrough time
- Wells located within the channel system, had the highest production rates because of the high permeabilities and thicknesses associated within this region
- Three (3) major production mechanisms were observed: maintenance of reservoir pressure, displacement of oil by nitrogen both horizontally and vertically and also the vaporization of the liquid hydrocarbon components from the oil column and possibly from the gas cap by nitrogen.

Synthetic Model

The results using the 25 acre standard model which tends to mimic onshore mature acreage conditions can be interpreted by the reader as per appendix 2. The discussion below though is relevant to both actual case and synthetic model.

Discussions

The results of the study indicate that the injection of nitrogen into the secondary gas cap is the optimum injection strategy to enhance oil recovery. The primary physical mechanisms that occurred as a result of nitrogen injection were:

- Partial maintenance of reservoir pressure.
The injection of nitrogen gas into the reservoir helps provides energy for maintaining the reservoir pressure to sustain oil production. By maintaining the reservoir pressure, the pressure difference between the reservoir and production wells remains high and hence oil production will also increase, as given by Darcy's law. Darcy's law states that the flow rate is directly proportional to pressure differential. In the simulation study, the low injection rates used was able to maintain the reservoir pressure at around 700 psi.
- Displacement of oil by nitrogen both horizontally and vertically.
For crestal or gas cap injection of nitrogen, the force of gravity tries to stabilize the downward nitrogen/oil displacement process by keeping nitrogen on top of the oil and counteracting the unstable nitrogen/oil viscous displacement process. As nitrogen is injected it invades the originally oil-saturated sand as the GOC moves downdip because of oil production farther downdip. The oil drains vertically downward through the nitrogen-invaded region and forms a thin layer with high oil saturation that drains along the base of the reservoir interval to the remaining downdip oil column.
- Vaporization of the liquid hydrocarbon components from the oil column and possibly from the gas cap by nitrogen.
Nitrogen injection is associated with mass transfer compositional effects. The injected nitrogen/oil composition interactions involve stripping effects i.e. various light components from the oil phase is transferred to the injected nitrogen.

From this study, several important concepts regarding immiscible nitrogen/oil displacements were learnt. These included:

- The immiscible nitrogen/oil displacement is generally an inefficient oil displacement process because nitrogen is a highly mobile fluid.
- The immiscible nitrogen/oil displacement process becomes efficient and desirable when gravity works to keep the low density nitrogen on top of the higher-density oil and when there is mass transfer of components from the oil to the nitrogen.
- The immiscible nitrogen/oil displacement process also becomes desirable when the depositional environment is indicative of an upward fining sequence, as this helps to reduce the upward migration of the less dense nitrogen gas to upper zones in lower dip environments.
- The most optimum strategy for nitrogen injection involves injecting nitrogen into the crestal secondary gas cap, with the oil wells producing from as far downdip as possible, by maximizing the distance from the gas cap both vertically and laterally.
- Nitrogen also has compositional mass transfer effects on the reservoir fluids. One nitrogen/oil compositional mass transfer effect is oil swelling. However, the effect in nitrogen is very small in this case as nitrogen has a very low solubility in oil as compared to other gases such as carbon dioxide.
- The other nitrogen/oil compositional mass transfer effect is stripping or vaporization of light hydrocarbon components from the oil by the nitrogen. Nitrogen is able to strip or vaporize the C₁-C₅ components which cause the density and viscosity of the oil to increase. This effect is very small and occurs with all types of oils but is more significant for lighter or higher API gravity oils.
- Upon contact with water present in the underlying aquifer, nitrogen has no effect on the water viscosity. However, an increase in water density is observed which is due primarily to the very low solubility of nitrogen in the water phase.

It is noteworthy, based on the author's experience, that more than 80 % of shallow onshore reservoirs that meet any criteria for IOR process possess dip between 15 degrees and 60 degrees. This is not only a significant factor for understanding gravity drainage in primary production but has significant effects on displacement from nitrogen or any other injection gas. Other points of interest include the following:

- Nitrogen can be more reliably supplied (on site production) than carbon dioxide (remote supply).
 - This fact reduces many infrastructure security concerns and production hiccups.
- Nitrogen does not have the health and safety issues on humans and environment as carbon dioxide.
 - Nitrogen breakthrough at surface sites due to faulting or degraded old well infrastructure does not have the detrimental effects as using carbon dioxide.
 - Nitrogen will not toxify fresh water aquifers.
- Nitrogen injection will attract any government incentives for EOR projects.
- Small scale nitrogen (immiscible) injection projects are very much economically viable.

Conclusions

From this study, the following conclusions were drawn:

- Nitrogen can be effectively used as an injection gas to help maximize oil recovery in Trinidad's onshore mature acreage with pilot project emphasis on dipping reservoirs.
- Oil recovery through gas injection in immiscible processes is more influenced by dip and crude gravity compared to porosity-permeability relationships.
- Crestal or gas gap injection is the most suitable nitrogen injection strategy for the dipping reservoirs with lighter crudes.
- Higher injection rates do not equate to higher cumulative recoveries.
- Incremental recoveries(%) using nitrogen injection for projects spanning 10yrs - 25 yrs is 10% to 21% based on dip, crude API and porosity-permeability relationship.

Appendix 1

OIL RECOVERIES FROM NITROGEN PROJECTS		
FIELD	EOR APPLICATION	RECOVERY (% OOIP)
Hawkins*	Gravity Drainage	20% *Carlson-ExxonMobil
Chunchula	Pressure Maintenance	30%
Block 31	Miscible Displacement	60% (Total, No Secondary)
Szeged Moraváros	Pressure Maintenance	12%
Jay/(LEC)	Miscible Displacement	13%
Lake Barre	Pressure Maintenance	19%
Fordoche Wilcox 8 & 12	Miscible Displacement	19.5%
Fordoche Wilcox 5	Pressure Maintenance	19% - 46%
Field B*	Immiscible Displacement	36% *Clancy
Anschutz Ranch East	Pressure Maintenance	45% - 53%

Table 1: Oil Recoveries from Past Nitrogen Injection Projects

TYPE	OPERATOR	POROSITY (%)	DEPTH (ft)	GRAVITY (°API)	OIL VISCOSITY (cP)	TOTAL PRODUCTION (BOPD)	EOR PRODUCTION (BOPD)
Immiscible	ExxonMobil	28	4,600	16	25	1,500	1,000
Immiscible	ExxonMobil	28	4,600	24	4	9,000	3,000
Immiscible	Occidental	28	2,500	25	20	20,000	5,000

Table 2: Recent Nitrogen Injection Projects

SUMMARY OF RESERVOIR ROCK & FLUID PROPERTIES: TRINIDAD EXAMPLE	
Lithology	Sandstone
Depositional Environment	Deepwater Turbidite Channel System
Depth (ft)	3,340-4,801
Thickness (ft)	10-110
Acres/Feet (ac/ft)	9,284
Initial Reservoir Pressure (psi)	1,950 @ 3807 ft
Current Reservoir Pressure (psi)	670
Porosity (%)	8-34
Permeability (md)	2-268
Dip (°)	25-30
Drive Mechanism	Solution Gas Drive, Weak Water Drive
Oil Gravity (°API)	35
Gas Gravity	0.8
Bubble Point Pressure (psi)	1,264

Table 3: Summary of Reservoir Rock & Fluid Properties - Trinidad Example

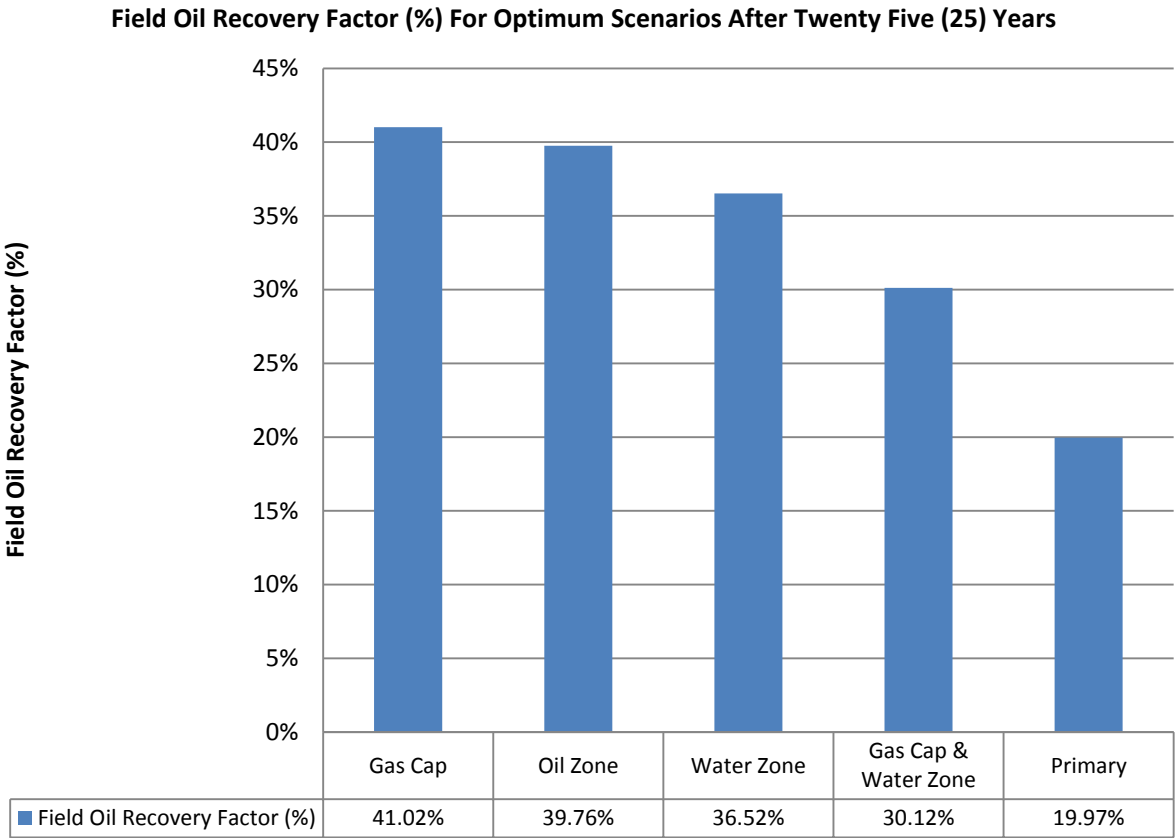


Figure 1: Results for Real Model, Trinidad

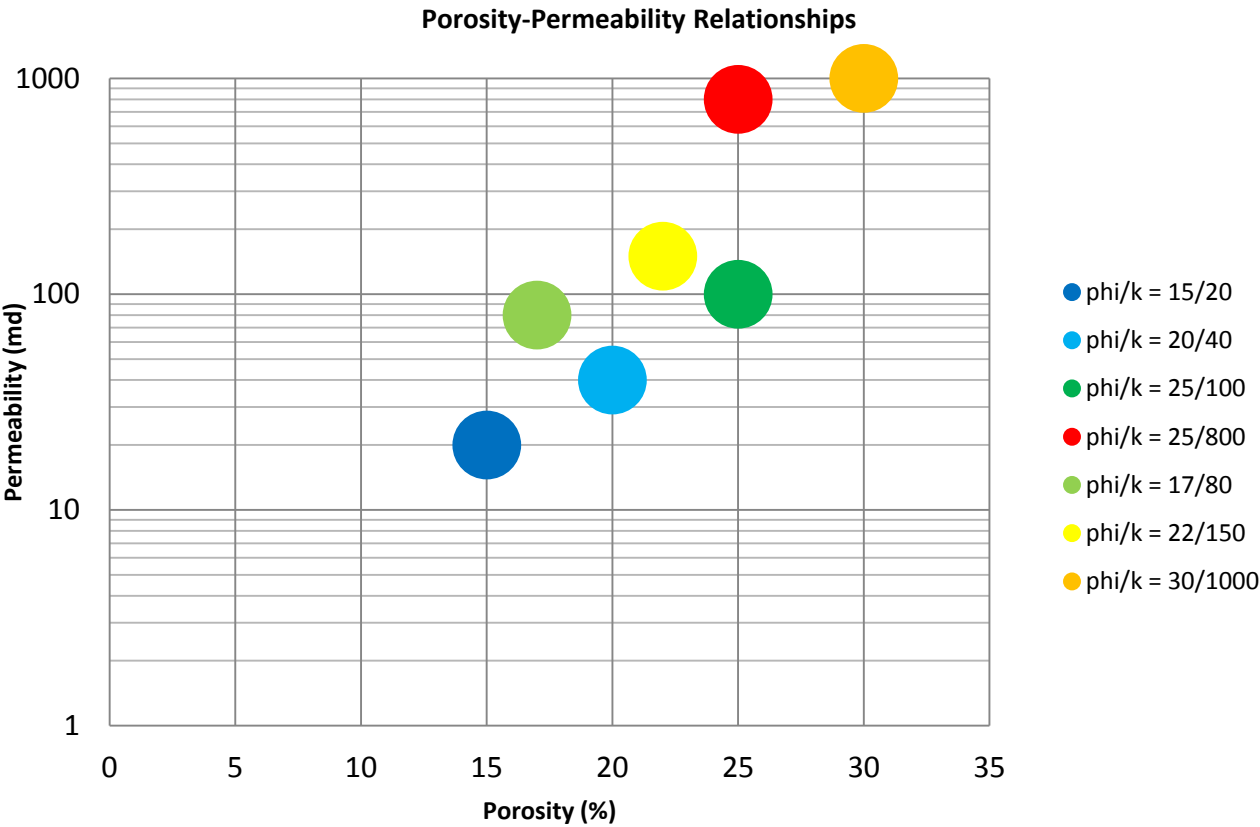
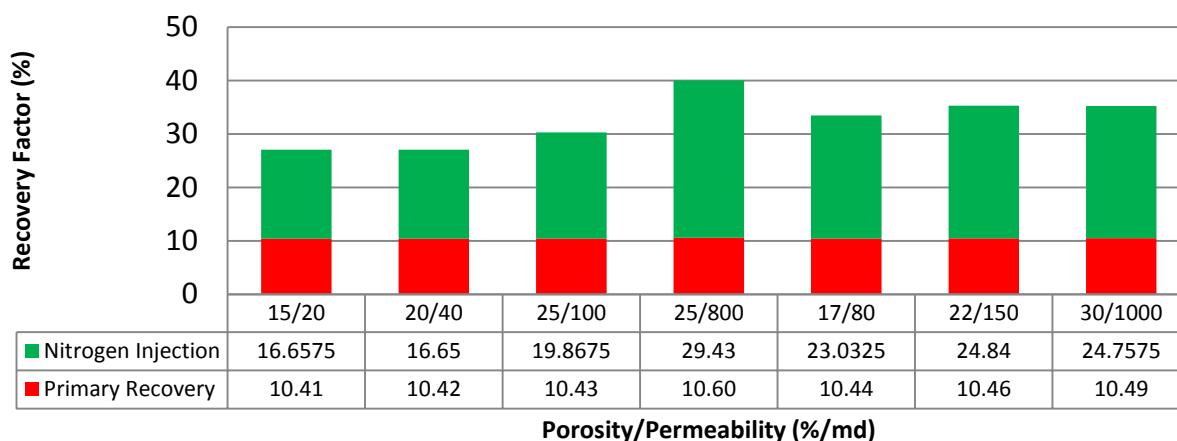


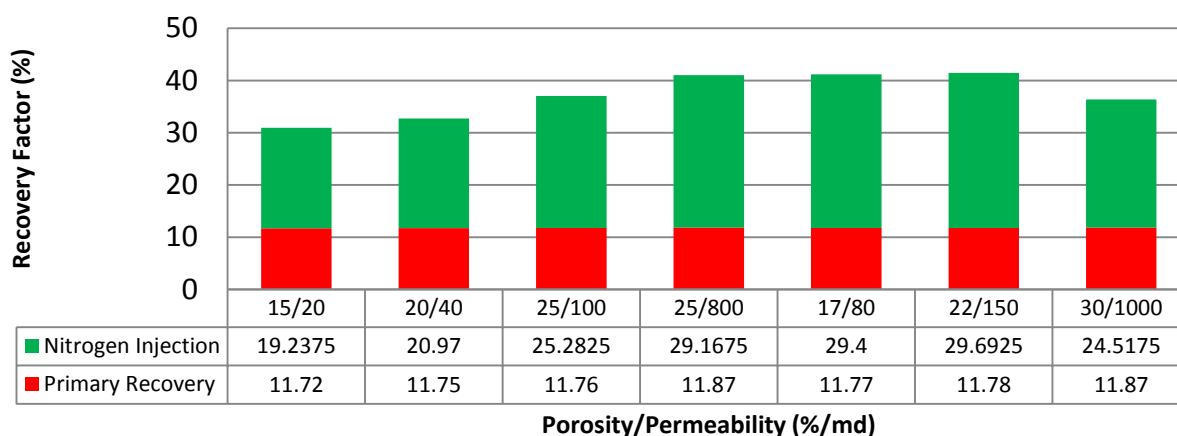
Figure 2: Porosity-Permeability Relationships

Appendix 2

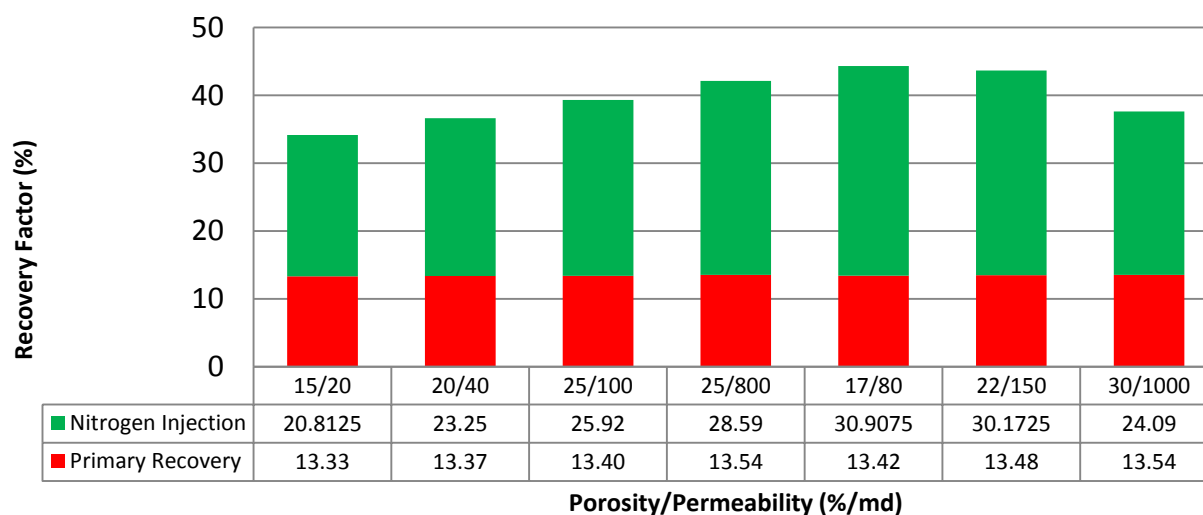
Recovery Factors for Scenario 1: 10 Degrees Formation Dip, 25 Degrees API Oil



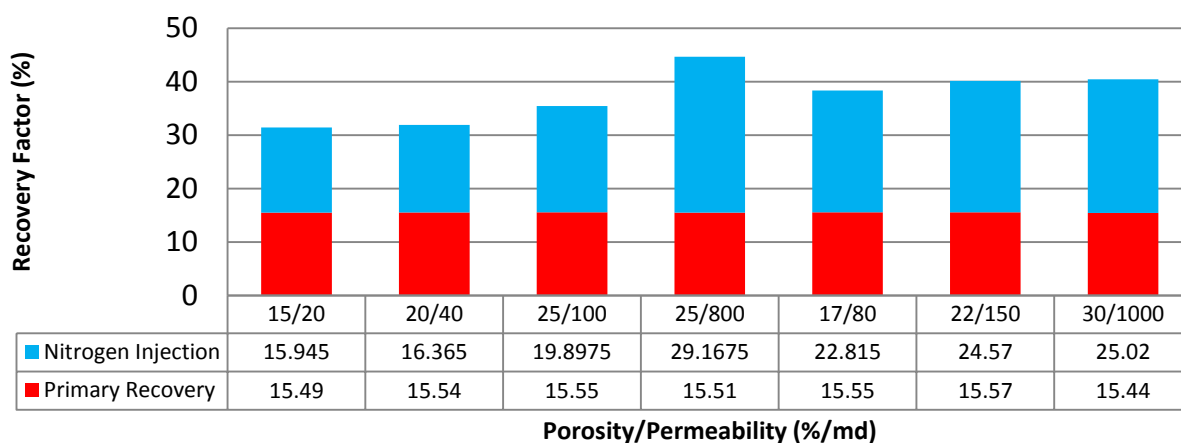
Recovery Factors For Scenario 2: 20 Degrees Formation Dip, 25 Degrees API Oil



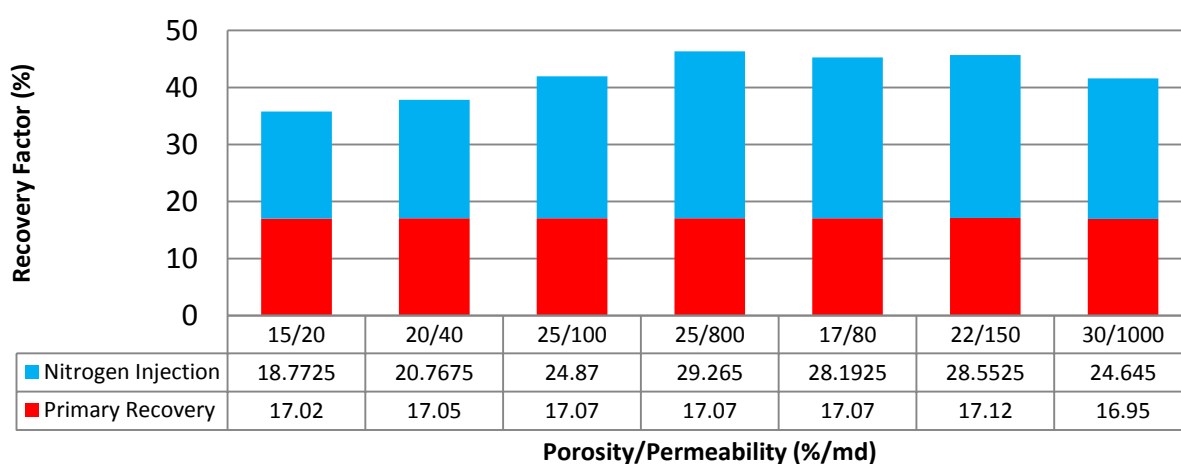
Recovery Factors For Scenario 3: 30 Degrees Formation Dip, 25 Degrees API Oil



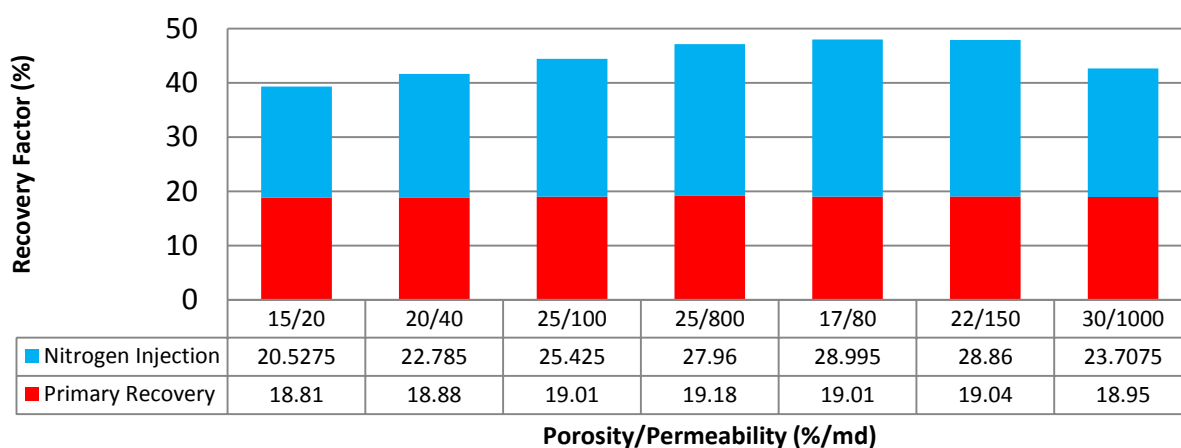
Recovery Factors For Scenario 4: 10 Degrees Formation Dip, 30 Degrees API Oil



Recovery Factors For Scenario 5: 20 Degrees Formation Dip, 30 Degrees API Oil



Recovery Factors For Scenario 6: 30 Degrees Formation Dip, 30 Degrees API Oil



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